E-COMMERCE ANALYTICAL REPORT

WEINTERN DATA SCIENCE INTERNSHIP — WEEK 1 FINAL SUBMISSION

CONFIDENTIAL / FINAL

1. TITLE AND OBJECTIVE

Project Title: End-to-End E-Commerce Data Cleaning, KPI Analytics & Reporting

Objective: Build a robust automated data preparation pipeline for raw transactional logs, eliminate missing and redundant entries, calculate core e-commerce metrics (KPIs), and present executive insights for business decisions.

2. DATASET OVERVIEW

Dimension	Details / Scope
Data Scope	Transactional record set featuring customer orders, prices, quantities, payment modes, and fulfillment statuses.
Core Features Tracked	`Order ID`, `Customer ID`, `Product Name`, `Category`, `Quantity`, `Unit Price`, `Total Amount`, `Order Date`, `Payment Mode`, `Delivery Status`, `City or Region`.

3. CLEANING SUMMARY

```
Raw Data ──▶ Drop Duplicates ──▶ Median Imputation ──▶ Formula Alignment ──▶ Datetime Cast ──▶ Cleaned CSV
```

- **Duplicates:** Screened and dropped redundant rows via `df.duplicated()` to ensure dataset uniqueness.
- **Missing Data Imputation:** Replaced missing *Quantity* and *Unit Price* using category-level median values. Recalculated *Total Amount* ($\text{Quantity} \times \text{Unit Price}$). Imputed missing *Payment Mode* and *Delivery Status* with mode values.
- **Data Standardisation:** Parsed string order dates into official `datetime64` for chronological trend plotting.
- **Output File:** Cleaned data stored at `data/cleaned/cleaned_ecommerce_data.csv`.

4. KPI SUMMARY

Key Performance Indicator	Measured Value	Executive Context
Total Revenue	₹67,300	Gross revenue realized across all processed orders.
Total Completed Orders	6	Total count of validated customer orders.
Average Order Value (AOV)	₹11,216.67	Strong average ticket size driven by high-end Electronics.
Top Revenue Category	Electronics	Drives over 74% of total transaction value.
Delivery Fulfillment Rate	66.7%	Delivered successfully (16.7% Pending, 16.7% Canceled).

5. TREND ANALYSIS & 6. CUSTOMER INSIGHTS

- **Sales Trends:** Major sales spikes occurred during early January 2026, primarily anchored by Electronics transactions.
- **Payment Mode Correlations:** High-ticket orders heavily lean on digital options (UPI / Credit Card), whereas routine items frequently use Cash on Delivery (COD).
- **Customer Profiles:** Customer C002 demonstrated repeat purchase behavior across Footwear and Apparel lines.
- **Regional Clusters:** Top revenue performance is concentrated in Tier-1 hubs like **Mumbai**, followed by secondary demand in **Pune** and **Nagpur**.

7. VISUALIZATION SECTION

All visual charts and dashboards are compiled and linked inside repository assets:

- **Category Revenue Bar Chart:** `outputs/charts/Screenshot_Bar_Chart.png`
- **Fulfillment Breakdown Pie Chart:** `outputs/charts/Screenshot_Pie_Chart.png`
- **Sales Trend Line Chart:** `outputs/charts/Screenshot_Line_Chart.png`
- **Executive Dashboard Mockup:** `dashboard/dashboard-mockup.png`

8. KEY FINDINGS

Strategic takeaway: Electronics yield 74%+ of sales volume. Pending and canceled logistics statuses correlate slightly with Cash-on-Delivery payment choices in specific transit channels.

9. RECOMMENDATIONS & 10. LIMITATIONS

Recommendations:

1. Focus inventory buffers for high-value Electronics and Footwear in Mumbai and Pune hubs.
2. Offer targeted cross-category promotional bundles to repeat customers (e.g. Customer C002).
3. Partner with reliable local logistics providers to push the 66.7% fulfillment benchmark higher.

Limitations & Assumptions: Based on a sample assignment dataset; reflects short-term January trends without long-term annual seasonality or promo discounts.